— *Solution notes* —

COMPUTER SCIENCE TRIPOS Part IB – 2021 – Paper 6

5 Computation Theory (amp12)

Universal register machine

(*a*) For each $n, e \in \mathbb{N}$, let $\varphi_e^{(n)}$ denote the partial function $\mathbb{N}^n \rightharpoonup \mathbb{N}$ computed by the register machine with index e using registers $\mathtt{R}_1, \ldots, \mathtt{R}_n$ to store the n arguments and register $\mathtt{R}_0$ to store the result, if any.

Explain why for each $m, n \in \mathbb{N}$ there is a totally defined register machine computable function $S_{m,n} : \mathbb{N}^{1+m} \to \mathbb{N}$ with the property that for all $(e, \vec{x}) \in \mathbb{N}^{1+m}$ and $\vec{y} \in \mathbb{N}^n$

$$\varphi^{(n)}_{S_{m,n}(e,\vec{x})}(\vec{y}) \equiv \varphi_e^{(m+n)}(\vec{x}, \vec{y}) \tag{1}$$

where $\equiv$ denotes Kleene equivalence: for all $z \in \mathbb{N}$, the left-hand side is defined and equal to z if and only if the right-hand side is defined and equal to z. Your explanation should make clear what assumptions you are making about the relationship between numbers and register machine programs. [10 marks]

Answer: We asume given effective operations for coding register machine (RM) programs P as numbers $\ulcorner P \urcorner$ and decoding numbers e as RM programs $prog(e)$. Then using the same techniques as for constructing a universal RM, one can construct a RM M to carry out the computation

$$\begin{array}{l} \textit{let } e = \mathtt{R}_1, x_1 = \mathtt{R}_2, \ldots, x_m = \mathtt{R}_{m+1} \textit{ in} \\ \mathtt{R}_0 ::= \ulcorner \mathtt{R}_{m+1} ::= \mathtt{R}_1; \cdots ; \mathtt{R}_{m+n} ::= \mathtt{R}_n; \mathtt{R}_1 ::= x_1; \cdots ; \mathtt{R}_m ::= x_m; \textit{prog}(e) \urcorner \end{array}$$

which always terminates. Starting with e in $\mathtt{R}_1$ and $\vec{x}$ in registers $\mathtt{R}_2, \ldots, \mathtt{R}_{m+1}$, this computes a value e' in $\mathtt{R}_0$ with the property that $prog(e')$ is a RM program that when started with $\vec{y}$ in $\mathtt{R}_1, \ldots, \mathtt{R}_n$, computes $\varphi_e^{(m+n)}(\vec{x}, \vec{y})$. Thus M computes a function $S_{m,n}$ satisfying (1); and since it always terminates, $S_{m,n}$ is a total RM computable function.

(*b*) Let $f : \mathbb{N}^{1+m+n} \rightharpoonup \mathbb{N}$ be a register machine computable partial function of $1+m+n$ arguments for some $m, n \in \mathbb{N}$.

(*i*) Why is the partial function $\hat{f} : \mathbb{N}^{1+m+n} \rightharpoonup \mathbb{N}$ satisfying for all $(z, \vec{x}, \vec{y}) \in \mathbb{N}^{1+m+n}$

$$\hat{f}(z, \vec{x}, \vec{y}) \equiv f(S_{1+m,n}(z, z, \vec{x}), \vec{x}, \vec{y}) \tag{2}$$

register machine computable? [3 marks]

Answer: Register machine computable partial functions are closed under composition; and projection functions are register machine computable. The partial function $\hat{f}$ is obtained by composing f (assumed to be computable), $S_{1+m,n}$ (computable by part (*a*)) and projections.

(*ii*) By considering $S_{1+m,n}(e, e, \vec{x})$ where e is an index for the partial function $\hat{f}$ in part (*b*)(*i*), prove that there is a totally-defined register machine computable function $\mathsf{fix}\, f : \mathbb{N}^m \to \mathbb{N}$ with the property that for all $\vec{x} \in \mathbb{N}^m$ and $\vec{y} \in \mathbb{N}^n$

$$\varphi^{(n)}_{\mathsf{fix}\, f(\vec{x})}(\vec{y}) \equiv f(\mathsf{fix}\, f(\vec{x}), \vec{x}, \vec{y}) \tag{3}$$

[7 marks]

Answer: Let $e \in \mathbb{N}$ be an index for $\hat{f}$. Thus

$$\varphi_e^{(1+m+n)}(z, \vec{x}, \vec{y}) \equiv f(S_{1+m,n}(z, z, \vec{x}), \vec{x}, \vec{y}) \tag{4}$$

holds for all $(z, \vec{x}, \vec{y}) \in \mathbb{N}^{1+m+n}$. For each $\vec{x} \in \mathbb{N}^m$ define

$$\mathsf{fix}\, f(\vec{x}) \triangleq S_{1+m,n}(e, e, \vec{x}) \tag{5}$$

This is a register machine computable total function $\mathbb{N}^m \to \mathbb{N}$, because it is obtained by composing the computable total function $S_{1+m,n}$ with other such functions (constant and identity functions). Furthermore, for all $(\vec{x}, \vec{y}) \in \mathbb{N}^{m+n}$ we have:

$$\begin{aligned}
\varphi^{(n)}_{\mathsf{fix}\, f(\vec{x})}(\vec{y}) &\equiv \varphi^{(n)}_{S_{1+m,n}(e,e,\vec{x})}(\vec{y}) && \text{by (5)} \\
&\equiv \varphi_e^{(1+m+n)}(e, \vec{x}, \vec{y}) && \text{by (1)} \\
&\equiv f(S_{1+m,n}(e, e, \vec{x}), \vec{x}, \vec{y}) && \text{by (4)} \\
&\equiv f(\mathsf{fix}\, f(\vec{x}), \vec{x}, \vec{y}) && \text{by (5)}
\end{aligned}$$

as required.
